K-SAE Analysis on Chinese Misty Poems

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Abstract

Chinese Misty poetry has long been regarded as difficult to analyze systematically due to its semantic ambiguity. This paper explores whether the semantic ambiguity in obscure poetry has a reproducible underlying structure and attempts to characterize it using digital methods. First, we employ a Chinese semantic embedding model based on Transformer to conduct representation learning on the poetry texts. Subsequently, we use the k-sparse autoencoder (k-SAE) to decompose the dense representations into sparse latent semantic features. On this basis, this paper introduces an interpretation and verification framework assisted by generative artificial intelligence to perform semantic annotation and consistency evaluation on the latent features. The experimental results show that some latent features exhibit stable activation patterns and high interpretability, indicating the existence of identifiable latent semantic structures in obscure poetry texts.

1 Introduction

1.1 Background

Poetry is often regarded as a literary form that is difficult to define. It rejects single answers, accommodates multiple interpretations, and often avoids a clear and stable meaning structure. This is particularly true of China's "Misty Poetry." According to Yeh (2016), since its inception, Misty Poetry has consistently been described as a genre of poetry that is obscure, ambiguous, and defies exhaustive comprehension. These poems are frequently criticized in the Chinese literary circle, and even their name is derogatory, originating from the incomprehensible nature of their text content. The issue this paper focuses on is precisely this long - standing implicit premise: Does the "incomprehensibility" of Misty Poetry necessarily imply its unanalyzability? In other words, is the ambiguity of Misty Poetry entirely individualized and random, or are there some common semantic features? However, Brooks's (1947) research posits that all poetry can be interpreted in terms of concepts like "imaginary" and "structure." Then, can the methods of digital humanities be used to explore the patterns or structures of these Misty Poems?

1.2 Related Works

This question holds significant importance in both the fields of literary research and digital humanities. There is extremely limited research on Misty Poetry, and existing studies mainly understand it through historical and interpretive approaches. Cai's (2019) research places political, cultural, and identity reconstruction, emphasizing its significance as a historical product and regarding "obscurity" as a response to traumatic experiences and ideological oppression. Within this framework, the ambiguity of Misty Poetry is considered an unformulatable aesthetic feature that requires understanding through historical narratives and theoretical interpretations. Misty Poetry emerged after the end of the formulaic literature and the Great Cultural Revolution in China. It re - established the legitimacy of individual experiences and the lyrical subject through symbolism, metaphor, and emotional language. However, this emphasis on ambiguity and uncertainty has also deterred many traditional studies. Nevertheless, Matthews et al. (2025) conducted a quantitative study on the semantic evolution process of language concepts in French using the k-sparse autoencoder (k-SAE) method, providing new ideas for this paper to analyze Misty Poetry using language models.

1.3 Hypothesis & Purpose

This paper attempts to pose new questions based on this research context. By using the k-SAE to extract potential features from the corpus of Misty Poetry and relying on large - scale language models to explain these features, this paper views poetry

as a linguistic object with multiple layers of meaning rather than an unanalyzable whole. This paper anticipates that Misty Poetry is not a random semantic collection but contains a set of potential semantics that repeatedly appear among different poets. Moreover, another objective of this paper is to explore these potential semantic features.

2 Data

The corpus are extracted manually from the "Chinese poetry library" (zhong hua shi ci ku) website. It is consisted of over 500 classic misty poetries written by representative poets such as Bei Dao, Shu Ting, Gu Cheng, and Mang Ke. Each poem was manually cleaned and standardized to retain only poet name, poem title, and poem text. All the poem sets are devided into more individual poems by subtittles. For the usage of embedding models, this projecy merged all the poem content into one tensor as an input.

3 Methods

3.1 Overview

Our methods mainly followed the steps of Matthews et al.'s (2025) research. First the paper use transformer to get the embedding vector of our corpus. Then this paper use k-SAE to sparse it. Given by the activation data and the samples, we prompt the Generative AI to interpret and verify the lantent features. At the end, we make an analysis on the features. Here is the work-flow as Figure 1.

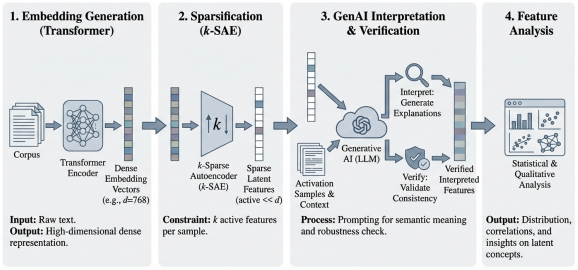


Figure 1: Proposed workflow for interpreting and verifying latent features using k-SAE and Generative AI.

Figure 1: Workflow

3.2 Embedding models

In the text representation stage, this paper uses the Chinese semantic embedding model `BAAI/bge-base-zh-v1.5` to map the original text into continuous vector representations. This model, based on the Transformer architecture, is trained on large-scale Chinese corpora and is capable of effectively capturing the contextual semantic information of the text, generating fixed-dimensional sentence-level embedding vectors. Its stable performance in the Chinese semantic representation task makes it suitable as the input representation for subsequent representation analysis methods.

After generating the embeddings, all vectors undergo standardization processing in the feature dimension to reduce the influence of differences in numerical scales across different dimensions on the model's training. The standardized embeddings are used as the input for k-SAE and are employed for subsequent sparse representation learning and latent feature analysis.

3.3 k-Sparse Autoencoders

This paper employs the k-Sparse Autoencoder (k-SAE) to conduct structured representation learning on high-dimensional semantic embeddings. The fundamental concept of k-SAE is to explicitly impose sparsity constraints in the latent space, ensuring that each input sample can only activate a limited number of latent dimensions. Through this constraint, the model can decompose the originally dense embedding representation into a set of relatively stable and distinguishable latent features, thereby providing a foundation for subsequent structural analysis and semantic interpretation. Compared with unconstrained autoencoder models, this sparse mechanism significantly reduces the co-activation phenomenon among latent features, making it easier for individual latent dimensions to correspond to consistent semantic or pragmatic patterns.

Formally, let the input embedding vector be $\mathbf{x} \in \mathbb{R}^d$, where d represents the embedding dimension. The encoder first maps the input to the latent space through a linear transformation, obtaining the latent pre - activation vector:

$$z = W_e x + b_e \tag{1}$$

Among them, $W_e \in \mathbb{R}^{n \times d}$ represents the encoder weight matrix, $b_e \in \mathbb{R}^n$ denotes the bias term, and n is the dimension of the latent space. Subsequently, a Top-k sparsification operation is applied to z, where only the k components with the largest values are retained, and the remaining components are set to zero, thereby obtaining the sparse latent representation:

$$h = \mathrm{TopK}(z). \tag{2}$$

This operation ensures that each sample activates at most k dimensions in the latent space. The decoder reconstructs the input based on this sparse representation, and its form is:

$$\hat{x} = W_d h + b_d, \tag{3}$$

where $W_d \in \mathbb{R}^{d \times n}$ and $b_d \in \mathbb{R}^d$. The model is trained by minimizing the mean squared error between the input vector and the reconstructed vector, and its reconstruction loss is defined as:

$$\mathcal{L}_{\text{rec}} = \mathbb{E}\left[\|x - \hat{x}\|_2^2\right]. \tag{4}$$

During the training process, to improve the overall utilization of the latent space and reduce the number of latent dimensions that have not been activated for a long time, an auxiliary loss term is introduced to encourage the modeling ability of latent features for reconstruction residuals. Considering both the reconstruction error and the utilization of the latent space, the final optimization objective is expressed as:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \alpha \mathcal{L}_{\text{aux}}, \tag{5}$$

where α is the weight coefficient of the auxiliary loss term. After the model training is completed, the stability and structural characteristics of latent features can be analyzed by the activation frequency of latent dimensions and the co - activation relationship between latent dimensions.

3.4 GenAI interpretor and predictor

During the explanation phase, the objective of this paper is to generate a readable and understandable natural language semantic description for each potential feature. Specifically, for each latent feature, we select two types of representative text samples from the original text dataset based on their activation intensity:

- Top activating samples: These are the texts where the activation value on a certain feature is on top, and they are used to reflect the semantic content corresponding to the activation of that feature.

- Zero-activating samples: These are the texts where the activation value on a certain feature is zero. They are used as a control to reflect the semantic content that is unrelated to that feature.

Subsequently, the above two sets of texts were input into a large-scale language model (here we used Qwen-7b due to the financial and time cost), and the model was guided to compare the differences between the two sets of texts, thereby summarizing the core semantic concept of "existing universally in non-zero activation samples but absent in zero activation samples". This process ultimately generated a concise natural language label (feature concept) for each latent feature, used to describe the semantic pattern captured by that feature.

In order to evaluate whether the generated feature semantic explanations are true, robust and capable of distinguishing, this paper further introduces a prediction verification stage based on generative artificial intelligence. The core objective of this stage is to verify whether the generated semantic descriptions can effectively distinguish between text samples that activate the feature and those that do not.

The specific methods are as follows. For each latent feature, this paper randomly selects a fixed number of text samples for verification, which includes:

- Several top activation samples, and set their true labels as +1.0;

- Several zero-activation samples were set, and their true labels were assigned as -1.0

Subsequently, this paper uses a computationally less expensive generative model (such as GPT-4o-mini) to take the "feature concept" and the "text to be judged" as inputs. The model is required to determine the degree of match between this text and the feature concept, and output a continuous confidence score, with the range being [-1,1][-1, 1]. Here, positive values indicate that the model considers the text to be semantically consistent with the feature, while negative values indicate a mismatch. The magnitude of the absolute value reflects the confidence level of the match. Based on the confidence scores output by the generative model and the true labels, this paper respectively calculates the classification accuracy (Accuracy), F1 score, and Pearson Correlation to evaluate the consistency between the prediction results of the generative model and the true activation states. If a certain latent feature shows high consistency in all of the above indicators(here it means all of them are larger than 0.8), it indicates that the semantic interpretation has strong verifiability and credibility.

3.5 Latent Feature Relationship Analysis

To examine whether interpretable latent features exhibit an observable organizational structure at the corpus level, we regard each screened latent feature as a network node and construct a feature network based on their co - activation relationships. Let the sparse activation matrix be $H \in \mathbb{R}^{T \times d}$. For any feature i, we define its set of activated samples as:

$$S_i = \{\, t \mid H_{t,i} \neq 0 \,\}.$$

Since the activation of k-SAE latent features is typically highly sparse and the "subset" phenomenon often occurs (more specific features are usually subsumed by more macroscopic features), using the Jaccard similarity

$$J(i,j) = \frac{|S_i \cap S_j|}{|S_i \cup S_j|}$$

will underestimate such subset - style co - occurrence relationships due to the dominance of the union term. To more effectively capture stable co - occurrence and inclusion relationships in sparse sets, we adopt the overlap coefficient:

$$O(i,j) = \frac{|S_i \cap S_j|}{\min\left(|S_i|, |S_j|\right)}.$$

This metric indicates the proportion of samples in the feature with fewer activated samples that also activate the other feature, and is more suitable for revealing the hierarchical and complementary structures of latent features.

Based on this, we construct a network $G = (V, E)$, where V consists of the latent features that have passed the interpretability screening. An edge is connected when

$$(i,j) \in E \iff O(i,j) \geq \tau$$

The threshold τ represents "at least a τ proportion of samples in the smaller activation set are shared with another feature", which is used to retain stable co-activations and suppress accidental co - occurrences. This network structure can be used to determine whether interpretable features form local clusters (indicating co - occurrence and hierarchical relationships of themes/rhetorical mechanisms) or are mainly distributed in isolation (indicating that features tend to depict distinct poetic patterns), thus providing structured evidence for our hypothesis.

4 Results

4.1 k-SAE hyperparameter tuning

Here, since our corpus have only 551 samples of misty poem, it has limited features. This paper chose k in $[4, 8, 16]$ and dimension in $[\frac{1}{4}d, \frac{1}{2}d, d, 2d]$ as our potential combinations. We made an record on dead latent evaluation as Fig 2, reconstruction loss as Fig 3, and aux loss during training as Fig 4.

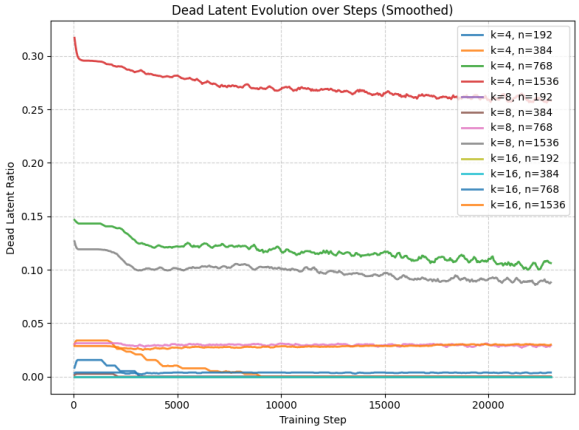


Figure 2: The Dead Latent Evoluation Over Steps

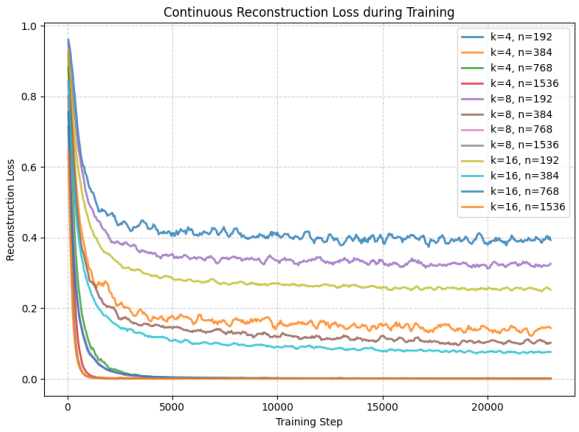


Figure 3: Continuous Reconstruction Loss during Training

Form these plots, we mainly focus on the training process and convergence. The most important comparison should be on feature density map. According to O'Neill et al. (2024), they provided an empirical range for interpretable features to be between -1 and -3 in scale of $\log_1 0$ while the peak lies in around -1.8.

The study also compared the feature density of the model under different combinations of the number of potential dimensions n and the sparsity parameter k, and needed to select the most likely

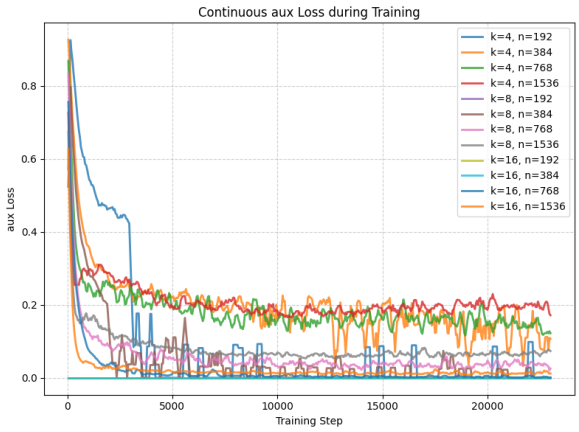


Figure 4: Continuous Aux Loss During Training



Figure 5: Latent Feature Relation Graph

combinations. Among these hyperparameter combinations, different values may also have very similar feature densities, but these combinations are not equivalent in terms of the interpretability of the latent features.

Specifically, when the number of potential dimensions n is fixed, a smaller sparsity parameter k enables each input sample to activate only a more limited set of potential features. This constraint can reduce the degree of co-activation among different potential features, minimize semantic overlap, and help form a more concentrated and semantically consistent potential representation. In contrast, a larger k, although it may maintain a similar activation density statistically, is more likely to cause a single sample to activate multiple potential features simultaneously, thereby weakening the semantic distinctiveness between the features.

On the other hand, when the sparsity parameter k is fixed, increasing the number of latent dimensions n can significantly enhance the capacity of the representation space, allowing the model to have more degrees of freedom to assign different semantic patterns to different latent features. This higher-dimensional latent space helps reduce semantic collisions (semantic collision), preventing multiple semantic patterns from being forced to map to the same latent feature, thereby improving the semantic specificity and interpretability of the latent features. Therefore, combining the empirical interval of the density function introduced earlier, we selected the combination with a smaller k and a larger n among the similar combinations that met the requirements. Eventually, there were three potential hyperparameter combinations, the data would be showed below.

4.2 interpretability

Here we interpret and make prediction on all three potential combination. Here is the comparison Table 1. From the Table, we can see the combination (k,n)=(4.384) have the highest interpretablity rate, so our final analysis is on this combination.

4.3 Feature Analysis

Based on the Latent feature relation Fig 5, The relationship graph exhibits a structure characterized by "overall sparsity and local clustering." The majority of interpretable features do not have significant co-activation among themselves, thus existing as isolated nodes. Only a few features form small connected components. This indicates that most of the latent features learned by the model correspond to relatively independent semantic or rhetorical triggering patterns, rather than a large number of highly redundant repetitive features. The visible edges in the graph mainly appear between features with similar semantics or those that can be regarded as hierarchically inclusive. For example, a group of features centered around "natural imagery—time/instantaneous feelings" form a chain connection (such as natural imagery, the transient beauty of nature, and time reflection), suggesting that these features are often co-triggered in the same type of text and may jointly depict the typical "image clusters" and temporal writing in misty poetry. Another group of edges connects "surreal urban—industrial imagery" and "surreal philosophical abstract imagery," indicating that abstract philosophical thoughts and surreal descriptions of cities/industries co-occur stably in some

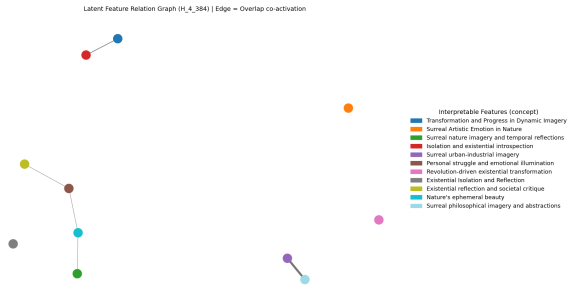


Figure 6: Latent feature relation graph (edges defined by overlap-based co-activation). Each node denotes a highly interpretable latent feature (color-coded; feature names listed in the legend). An undirected edge is drawn between two features when their overlap coefficient exceeds 0.6, indicating that a large fraction of the smaller feature's activation set co-occurs with the other; edge width scales with overlap strength.

samples. In contrast, nodes such as "Surreal Artistic Emotion in Nature" and "Revolution-driven existential transformation" remain isolated, meaning that they capture more distinguishable text patterns with more specific triggering conditions. Overall, the graph provides structural evidence for our hypothesis: the interpretable latent features of misty poetry are not random noise, but rather exhibit local associations centered around themes/images and a certain degree of hierarchy, while retaining a large number of distinct independent semantic dimensions, reflecting the "polysemous but organized" feature space of misty poetry.

5 Discussion

The core question of this paper is: Does the "incomprehensibility" of misty poems imply that they are "unanalyzable" in terms of structure? Our hypothesis is that obscurity is not a purely random collection of semantics but rather involves a repeatable underlying semantic mechanism. The overall results support this hypothesis. Through Chinese semantic embedding and sparse decomposition of k - SAE, we obtained a set of latent features that can be stably named. More importantly, these features achieved high consistency (high F1 score, accuracy, and correlation coefficient) in the subsequent validation of generative model prediction, indicating that these features are not only "describable" but also "testable," and their activation is not accidental noise.

This implies that obscurity and structure can co - exist. A single poem still allows for multiple interpretations, yet there are patterns that recur across texts at the semantic mechanism level (such as natural imagery combined with temporal reflection, existential introspection, surreal urban/industrial imagery associated with philosophical abstraction, etc.). Additionally, the co - activation relationship graph presents a structure of "overall sparsity and local clustering": most features are independent of each other, while a few form small associated clusters, suggesting that the model captures multiple relatively independent poetic mechanisms and their local co-occurrence/hierarchical relationships. In conclusion, the finding of this paper is that misty poems are not unanalyzable wholes, and their "ambiguity" can be understood as a structural phenomenon composed of a set of reproducible and interpretable underlying semantic dimensions.

6 Limitations

The core objective of this research is not to provide a definitive interpretation of the meaning of poems but to uncover the potential structural patterns in misty poems. However, it should be emphasized that the interpretable latent features do not equate to the meaning of the poems themselves. Even if the K-SAE can extract a stable semantic-emotional structure, the LLM's interpretation of these features remains an approximate linguistic expression rather than a final verdict on the meaning of the poems. The polysemy, openness, and aesthetic experience of poems cannot be fully encompassed by any single model or set of features.

Secondly, large language models are inherently model-dependent. Different corpora and different prompt designs will yield different results. Therefore, the LLM should be regarded as a heuristic tool.

Thirdly, the stability of latent features largely depends on the quality and quantity of the corpus. Here, although we have collected all the misty poems from the Chinese Poetry Database, due to the often-criticized nature of misty poems, there is not enough publicly available poem data to support in-depth research. Moreover, its greatest limitation is the inability to connect it with history, as we have only studied the misty poems themselves. If other literary works from the same period can be studied simultaneously and the connections between misty poems and them can be found, more profound conclusions may be drawn.

Table 1

	n	k	Dead Latents	Mean Acc.	Mean F1	Mean Corr.	Highly Interp.	% Highly Interp.
0	384	4	0	0.5015	0.6099	0.0703	11	2.86
1	384	8	0	0.5134	0.6137	0.0687	10	2.60
2	768	16	3	0.4975	0.6095	0.0619	15	1.95

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